SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	K Suthesika	**Reg. No.:**	192511070
Section / Batch:	BE CSE	**Date:**	24/07/2026

Code of Conduct

I , K Suthesika (192511070) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Suthesika

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

SET - 1

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q1 **Binary Search** 5 marks

Identify the error in the following snippet.

```
int binarySearch(int arr[], int n, int key) {
int low = 0, high = n - 1;
while(low < high) {
int mid = (low + high) / 2;
if(arr[mid] == key)
return mid;
else if(arr[mid] < key)
low = mid;
else
high = mid;
}
return -1;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int n, int key) {
    int low = 0, high = n - 1;

    while (low <= high) {
        int mid = (low + high) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}
int main() {
    int arr[] = {10,20,30,40,50};
    int n = 5;
    int key = 30;

    int result = binarySearch(arr,n,key);
    if (result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Element found at index 2
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q2 Stack 5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
if(top == 0) {
printf("Underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

int pop() {
    if (top == -1) {
        printf("Stack Underflow\n");
        return -1;
    }
    return stack[top--];
}
int main() {
    stack[++top]=10;
    stack[++top]=20;
    stack[++top]=30;

    printf("Deleted element: %d\n",pop());
    printf("Deleted element: %d\n",pop());
    printf("Deleted element: %d\n",pop());
    printf("Deleted element: %d\n",pop());

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Deleted element: 30
Deleted element: 20
Deleted element: 10
Stack Underflow
Deleted element: -1
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q1 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

int stack[5];
int top;

void init() {
    top = -1;
}
int main() {
    init();
    printf("Top = %d\n", top);
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Top = -1
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {
while(stack[top] != '(') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 100

char stack[MAX];
char postfix[MAX];
int top = -1;
int j = 0;

void push(char ch) {
    stack[++top] = ch;
}

char pop() {
    if (top == -1)
        return '\0';
    return stack[top--];
}

int main() {
    char ch = ')';

    push('(');
    push('+');
    push('*');

    if (ch == ')') {
        while (top != -1 && stack[top] != '(') {
            postfix[j++] = pop();
        }
        if (top != -1)
            pop();
    }
    postfix[j] = '\0';

    printf("Postfix: %s\n", postfix);

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
Postfix: *+
```

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QUESTIONS

Q1 Binary Search 5 marks
Q2 Stack 5 marks
Q3 Stack 5 marks
Q4 Stack 5 marks
Q5 Stack - Balancing Symbols 5 marks
Q6 Singly Linked List 5 marks
Q7 Stack 5 marks
Q8 Linked List 5 marks
Q9 Linked List 5 marks
Q10 Queue 5 marks

10/10 answered

Q5 **Stack - Balancing Symbols** 5 marks

In balancing Bracket, you identify the error.
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
pop();
} else {
return 0;
}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 100

char stack[MAX];
int top = -1;

void push(char ch) {
    stack[++top] = ch;
}
char pop() {
    if (top == -1)
        return '\0';
    return stack[top--];
}

int main() {
    char expr[] = "{[()]}";
    int i = 0;

    while (expr[i] != '\0') {
        char ch = expr[i];

        if (ch == '(' || ch == '[' || ch == '{') {
            push(ch);
        }
        else if (ch == ')' || ch ==']' || ch == '}') {
            if (top == -1) {
                printf ("Not Balanced\n");
                return 0;
            }
            if ((ch == ')' && stack[top] == '(') ||
                (ch == ']' && stack[top] == '[') ||
                (ch == '}' && stack[top] == '{')) {
                pop();
            }
            else {
                printf ("Not Balanced\n");
                return 0;
            }
        }

        i++;
    }
    if (top == -1)
        printf("Balanced\n");
    else
        printf("Not Balanced\n");

    return 0;
}
```

Test Input

stdin input (optional)

Output

Balanced

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List 5 marks

Q9 Linked List 5 marks

Q10 Queue 5 marks

10/10 answered

Q6 **Singly Linked List** 5 marks

What is the issue?

temp = head

while temp.next != NULL:

print(temp.data)

temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

struct Node {
    int data;
    struct Node *next;
};

int main() {
    struct Node n1,n2,n3;
    struct Node *head, *temp;

    n1.data = 10;
    n2.data = 20;
    n3.data = 30;

    n1.next = &n2;
    n2.next = &n3;
    n3.next = NULL;

    head = &n1;

    temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
10 20 30
```

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if(top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }
    stack[++top] = value;
}
int pop()
{
    if(top == -1)
    {
        printf("Stack Underflow\n");
        return -1;
    }
    return stack[top--];
}
int main()
{
    int n,i,x;

    scanf("%d", &n);

    for(i=0; i<n; i++)
    {
        scanf("%d",&x);
        push(x);
    }
    printf("%d\n",pop());

    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

- Q1 Binary Search 5 marks
- Q2 Stack 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Stack - Balancing Symbols 5 marks
- Q6 Singly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Linked List 5 marks
- Q9 Linked List 5 marks
- Q10 Queue 5 marks

10/10 answered

Q8 **Linked List** 5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};
int main() {
    struct Node *head, *prev = NULL, *current, *next;

    struct Node *n1 = (struct Node *)malloc(sizeof(struct Node));
    struct Node *n2 = (struct Node *)malloc(sizeof(struct Node));
    struct Node *n3 = (struct Node *)malloc(sizeof(struct Node));

    n1->data = 10;
    n2->data = 20;
    n3->data = 30;

    n1->next = n2;
    n2->next = n3;
    n3->next = NULL;

    head = n1;
    current = head;

    while (current != NULL) {
        next = current->next;
        current->next = prev;
        prev = current;
        current = next;
    }
    head = prev;

    current = head;
    while (current != NULL) {
        printf("%d ", current->data);
        current = current->next;
    }

    return 0;
}
```

Test Input

stdin input (optional)

Output

```
30 20 10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q9 **Linked List** 5 marks

What is the error in the below code?

new.next = head

head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *next;
};

int main(){
    struct Node *head = NULL;
    struct Node *new;

    new = (struct Node *)malloc(sizeof(struct Node));

    new->data = 10;

    new->next = head;
    head = new;

    struct Node *temp = head;
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q10 **Queue**

5 marks

What is missing in the below pseudocode?
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int Q[MAX];
int front = -1, rear = -1;

void enqueue(int x) {
    if (rear == MAX - 1) {
        printf("Queue Overflow\n");
        return;
    }
    if (front == -1)
        front = 0;

    Q[++rear] = x;
}

int dequeue() {
    if (front == -1) {
        printf("Queue Underflow\n");
        return -1;
    }
    int item = Q[front++];

    if (front > rear) {
        front = rear = -1;
    }
    return item;
}
int main() {
    int n, i, x;

    scanf("%d", &n);

    for (i=0; i<n; i++) {
        scanf("%d", &x);
        enqueue(x);
    }
    printf("%d\n", dequeue());
    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
10
```

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SET – 2

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue · Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack · Postfix Expression
5 marks

10/10 answered

Q1 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if(top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }
    stack[++top] = value;
}
int peek()
{
    if(top == -1)
    {
        printf("Stack Underflow\n");
        return -1;
    }
    return stack[top];
}
int main()
{
    int n, i, x;

    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        scanf("%d",&x);
        push(x);
    }
    printf("%d", peek());

    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q2 Circular Queue

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;
queue[rear] = vehicle;
if(rear == front){
printf("Queue Full");
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int vehicle)
{
    if ((rear+1)% MAX == front)
    {
        printf("Queue Full\n");
        return;
    }
    if (front == -1)
        front = 0;
    rear = (rear+1) % MAX;
    queue[rear] = vehicle;
}
int main()
{
    int n, i, x;

    scanf("%d", &n);

    for (i=0;i<n;i++)
    {
        scanf("%d",&x);
        enqueue(x);
    }
    printf("Front = %d\n", queue[front]);
    printf("Rear = %d\n", queue[rear]);

    return 0;
}
```

Test Input

```
3
101
102
103
```

Output

```
Front = 101
Rear = 103
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q3 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if(top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }
    stack[++top] = value;
}
int main()
{
    int n, i, x;

    scanf("%d", &n);

    for (i=0;i<n;i++)
    {
        scanf("%d", &x);
        push(x);
    }
    if(top != -1)
        printf("%d\n", stack[top]);
    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List 5 marks

Q6 Queue 5 marks

Q7 Queue · Deque 5 marks

Q8 Binary Search 5 marks

Q9 Circular Queue 5 marks

Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int stack[MAX];
int top = -1;

void push(int value)
{
    if(top == MAX - 1)
    {
        printf("Stack Overflow\n");
        return;
    }
    stack[++top] = value;
}
int pop()
{
    if (top == -1)
    {
        printf("Stack Underflow\n");
        return -1;
    }
    return stack[top--];
}
int main()
{
    int n, i, x;
    scanf ("%d", &n);
    for(i=0;i<n;i++)
    {
        scanf("%d", &x);
        push(x);
    }
    printf("%d\n", pop());
    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
30
```

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CO2 AT2 DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Dequeue 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q5 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
if (del->prev != NULL)
del->prev->next = del->next;
if (del->next != NULL)
del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
    int data;
    struct node *prev;
    struct node *next;
};
void deletenode(struct node **head, struct node *del)
{
    if(del == NULL)
        return;
    if (*head == del)
        *head = del->next;
    if(del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;

    free(del);
}
void printlist(struct node *head)
{
    while (head != NULL)
    {
        printf("%d ", head->data);
        head = head->next;
    }
}
int main()
{
    struct node *head;
    struct node *n1 = (struct node *)malloc(sizeof(struct node));
    struct node *n2 = (struct node *)malloc(sizeof(struct node));
    struct node *n3 = (struct node *)malloc(sizeof(struct node));

    n1->data=10;
    n2->data=20;
    n3->data=30;

    n1->prev=NULL;
    n1->next=n2;

    n2->prev=n1;
    n2->next=n3;

    n3->prev=n2;
    n3->next=NULL;

    head=n1;

    deletenode(&head, n2);

    printlist(head);
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
10 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue · Deque
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack · Postfix Expression
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int data)
{
    if(rear == MAX - 1)
    {
        printf("Queue Overflow\n");
        return;
    }
    if (front == -1)
        front = 0;
    queue[++rear] = data;
}
int main()
{
    int n, i, x;
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        scanf("%d",&x);
        enqueue(x);
    }
    printf("Front = %d\n", queue[front]);
    printf("Rear = %d\n", queue[rear]);

    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
Front = 10
Rear = 30
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

Q7 **Queue - Deque** 5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {
if (front == 0) front = MAX - 1;
else front = front - 1;
deque[front] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>

#define MAX 5

int dequeue[MAX];
int front = -1;
int rear = -1;

void insertFront(int data)
{
    if((front == 0 && rear == MAX - 1) || (front == rear +1))
    {
        printf("Dequeue Overflow\n");
        return;
    }
    if (front == -1)
    {
        front = rear = 0;
    }
    else if (front == 0)
    {
        front = MAX - 1;
    }
    dequeue[front] = data;
}
int main()
{
    int n, i, x;
    scanf("%d",&n);
    for(i=0;i<n;i++)
    {
        scanf("%d", &x);
        insertFront(x);
    }
    printf("Front Element = %d\n", dequeue[front]);
    return 0;
}
```

Test Input

```
3
10
20
30
```

Output

```
Front Element = 30
```

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue · Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack · Postfix Expression 5 marks

10/10 answered

Q8 **Binary Search** 5 marks

Identify the bug in this Recursive Binary Search.

int binarySearch(int arr[], int l, int r, int x) {

int mid = l + (r - l) / 2;

if (arr[mid] == x) return mid;

if (arr[mid] > x) return binarySearch(arr, l, mid, x);

return binarySearch(arr, mid, r, x);

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

int binarySearch(int arr[], int l, int r, int x)
{
    if(l>r)
        return -1;
    int mid = l + (r - l) / 2;

    if(arr[mid] == x)
        return mid;

    if (arr[mid] > x)
        return binarySearch(arr, l, mid - 1, x);
    return binarySearch(arr, mid + 1, r, x);
}
int main()
{
    int n, i, x;

    scanf("%d",&n);
    int arr[n];

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);
    scanf("%d", &x);

    int result = binarySearch(arr, 0, n-1, x);

    if(result == -1)
        printf("Element not found");
    else
        printf("%d",result);
    return 0;
}
```

Test Input

```
5
10 20 30 40 50
40
```

Output

```
3
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q9 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```c
#include <stdio.h>

#define max 5

int queue[max];
int front = -1, rear = -1;

void enqueue(int data)
{
    if ((rear + 1) % max == front)
    {
        printf("Queue Full\n");
        return;
    }
    if (front == -1)
        front = rear = 0;
    else
        rear = (rear + 1) % max;
    queue[rear] = data;
}
int dequeue()
{
    if (front == -1)
        return -1;
    int data = queue[front];

    if (front == rear)
    {
        front = rear = -1;
    }
    else
    {
        front = (front + 1) % max;
    }
    return data;
}
int main()
{
    int n, i, x;
    scanf("%d", &n);

    for(i=0;i<n;i++)
    {
        scanf("%d",&x);
        enqueue(x);
    }
    printf("%d\n", dequeue());

    return 0;
}
```

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Test Input

```
3
10
20
30
```

Output

```
10
```

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

- Q1 Stack 5 marks
- Q2 Circular Queue 5 marks
- Q3 Stack 5 marks
- Q4 Stack 5 marks
- Q5 Doubly Linked List 5 marks
- Q6 Queue 5 marks
- Q7 Queue - Deque 5 marks
- Q8 Binary Search 5 marks
- Q9 Circular Queue 5 marks
- Q10 Stack - Postfix Expression 5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {
if (isdigit(expression[i])) push(expression[i] - '0');
else {
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
#include <stdio.h>
#include <ctype.h>

int stack[100], top = -1;

void push(int x) {
    stack[++top] = x;
}
int pop() {
    return stack[top--];
}
int main() {
    char expression[] = "23+5*";
    int i = 0;

    while (expression[i] != '\0') {
        if(isdigit(expression[i])) {
            push(expression[i] - '0');
        } else {
            int op1 = pop();
            int op2 = pop();

            if (expression[i] == '+')
                push(op2+op1);
            else if (expression[i] == '-')
                push(op2-op1);
            else if (expression[i] == '*')
                push(op2*op1);
            else if (expression[i] == '/')
                push(op2/op1);
        }
        i++;
    }
    printf("%d", pop());
    return 0;
}
```

Test Input

stdin input (optional)

Output

```
25
```

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